

Antenna matching tool for ST25R NFC/HF readers

Data brief

Features

- STSW-ST25R004 is a software that calculates the matching network for antennas connected to ST25R NFC/HF readers, using Qucs (Quite Universal Circuit Simulator) to simulate the antenna and the matching network characteristics
- The software supports the following products:
 - ST25R500/ST25R501
 - ST25R300
 - ST25RN300
 - ST25R210
 - ST25R100/ST25R200
 - ST25R3920/ST25R3920B
 - ST25R3919B
 - ST25R3918
 - ST25R3917/ST25R3917B
 - ST25R3916/ST25R3916B
 - ST25R3915
 - ST25R3914

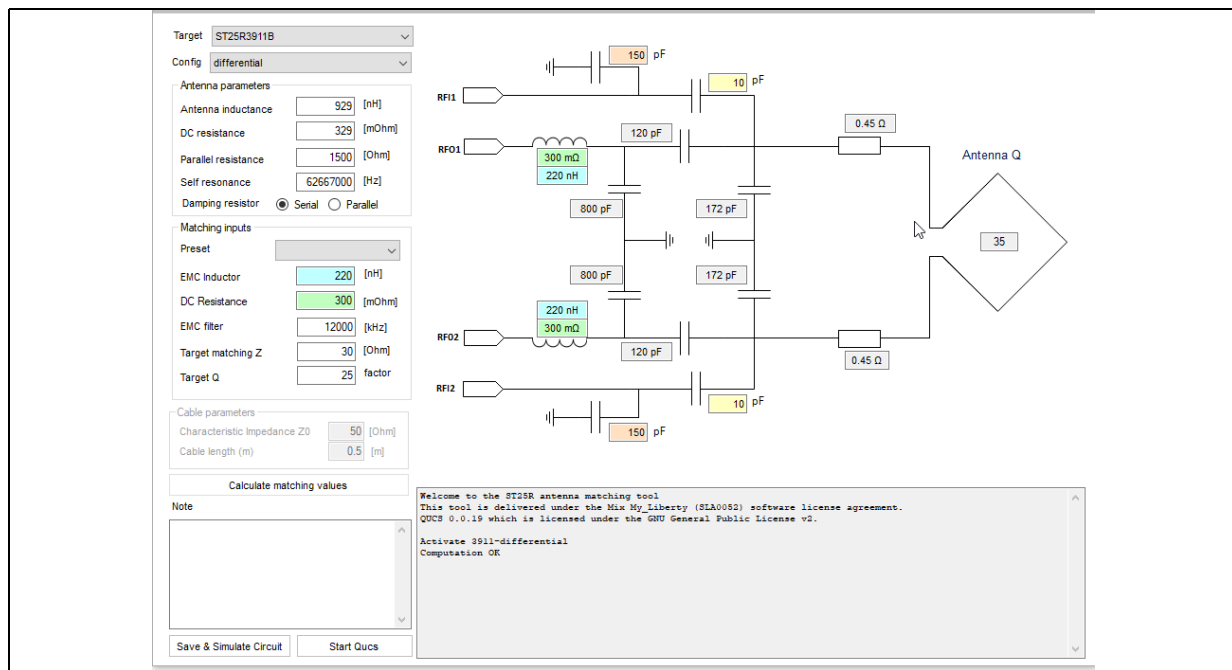
- ST25R3912
- ST25R3911B

Description

STSW-ST25R004 contains a graphical user interface (GUI) to specify the antenna parameters and the target matching. It performs transient, AC, and S-parameter simulations of the calculated antenna schematic.

An online version of the tool is available at <https://eds.st.com>. This is part of the ST eDesignSuite, an easy-to-use comprehensive software, helping customers during their design.

AN4974, AN5276, AN6065, and AN6092 (available on www.st.com) provide guidance on the measurement of the antenna parameters required as input for this matching tool, to define the best matching circuit.



1 License

STSW-ST25R004 is delivered under the Mix My_Liberty (SLA0052) software license agreement.

The software components provided in this package come with different license schemes, as shown in [Table 1](#).

Table 1. License scheme

Software component	Owner	License
ST25R Antenna Matching Tool binary	STMicroelectronics	ST proprietary
Qucs	Qucs team	GPL v2

2 Revision history

Table 2. Document revision history

Date	Revision	Changes
05-Jul-2017	1	Initial release.
19-Feb-2020	2	Updated document title, <i>Features</i> and <i>Description</i> . Added image on cover page and <i>Section 1: License</i> .
01-Jun-2021	3	Updated <i>Features</i> and <i>Description</i> .
24-Jun-2021	4	Updated <i>Description</i> .
03-May-2022	5	Updated document title, <i>Features</i> , and image on cover page.
26-Sep-2023	6	Updated <i>Features</i> .
17-Jul-2024	7	Updated document title, <i>Features</i> , and <i>Description</i> .
21-May-2025	8	Updated <i>Features</i> and <i>Description</i> .
23-Sep-2025	9	Updated <i>Features</i> and <i>Description</i> .
20-Feb-2026	10	Updated <i>Features</i> .

IMPORTANT NOTICE – READ CAREFULLY

STMicroelectronics NV and its subsidiaries ("ST") reserve the right to make changes, corrections, enhancements, modifications, and improvements to ST products and/or to this document at any time without notice.

In the event of any conflict between the provisions of this document and the provisions of any contractual arrangement in force between the purchasers and ST, the provisions of such contractual arrangement shall prevail.

The purchasers should obtain the latest relevant information on ST products before placing orders. ST products are sold pursuant to ST's terms and conditions of sale in place at the time of order acknowledgment.

The purchasers are solely responsible for the choice, selection, and use of ST products and ST assumes no liability for application assistance or the design of the purchasers' products.

No license, express or implied, to any intellectual property right is granted by ST herein.

Resale of ST products with provisions different from the information set forth herein shall void any warranty granted by ST for such product.

If the purchasers identify an ST product that meets their functional and performance requirements but that is not designated for the purchasers' market segment, the purchasers shall contact ST for more information.

ST and the ST logo are trademarks of ST. For additional information about ST trademarks, refer to www.st.com/trademarks. All other product or service names are the property of their respective owners.

Information in this document supersedes and replaces information previously supplied in any prior versions of this document.

© 2026 STMicroelectronics – All rights reserved